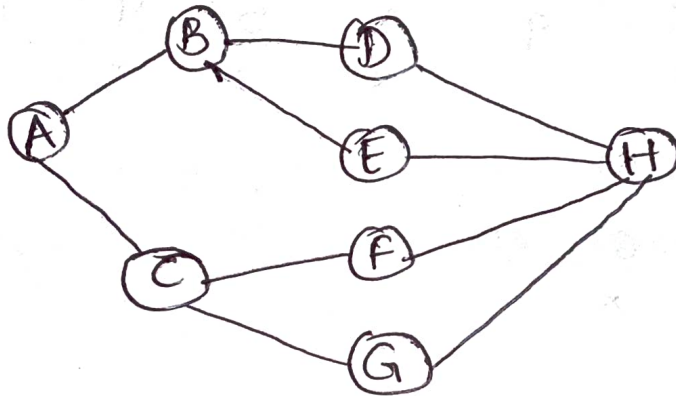


1. Explain the process of evaluating the postfix expression and evaluate the expression

$$734 + - 245 / + * 6 / 7 +$$

2. Explain BFS and DFS method for the following graph using start vertex as A



3. Construct Binary Tree by considering the following tree traversals

Inorder: N Z V A M C B S X D

Postorder: Z A V N C S D X B M

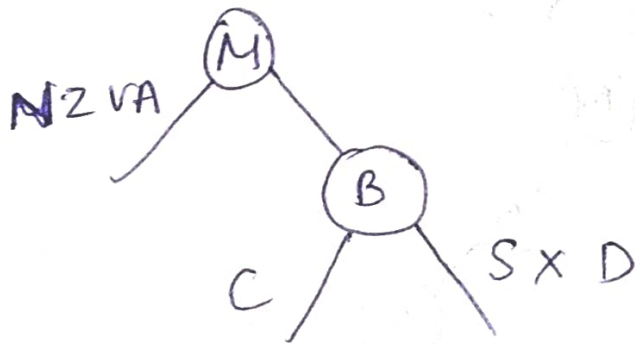
Answers:

3. Step 1: Take the last node in the post order that is M. The M is the root node.

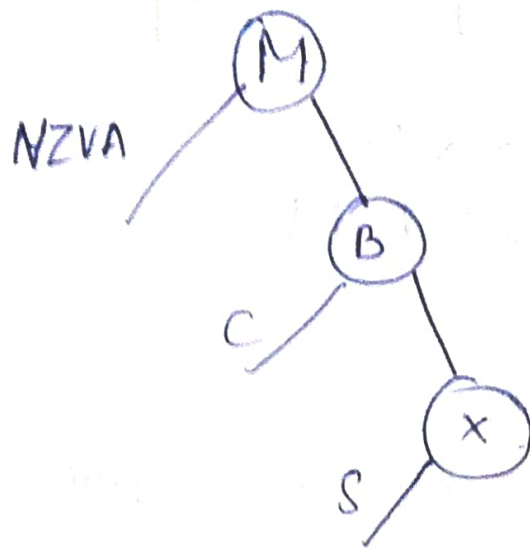


Now see the M in the inorder. The left side nodes of M are left subtree and right side nodes of M are right subtree.

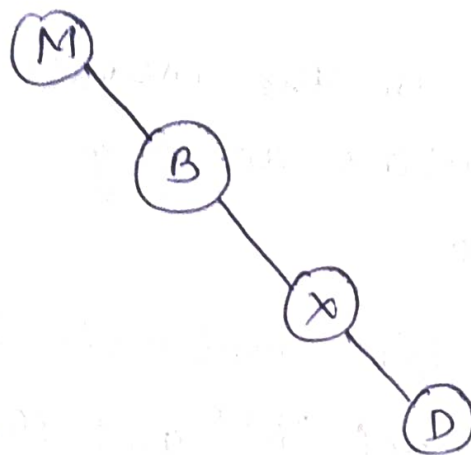
Step 2: The node before root node is B. It is the right child of root 'M' and the nodes left of B are left subtree of B and the nodes right of B are right subtree of B.



Step 3: The node before B in post order is X. The X is placed at right of B. and the nodes left of X is left subtree of X and the nodes right of X is right subtree of D.

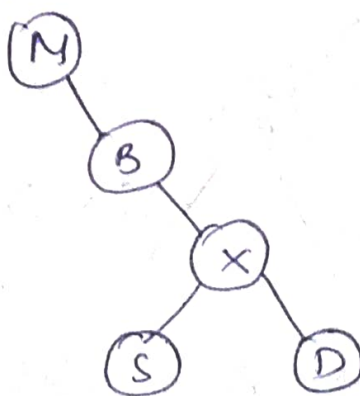


Step 4: The node before x in postorder is D. The D is placed at right of X and there is no nodes left and right of D.



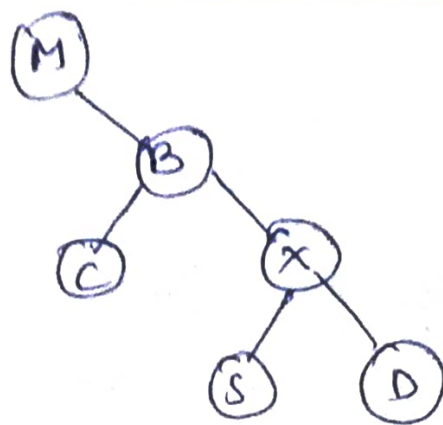
The D has no child. So left subtree of x is done.

Step 5: The node before D in the post order is S. S is the right node of X.



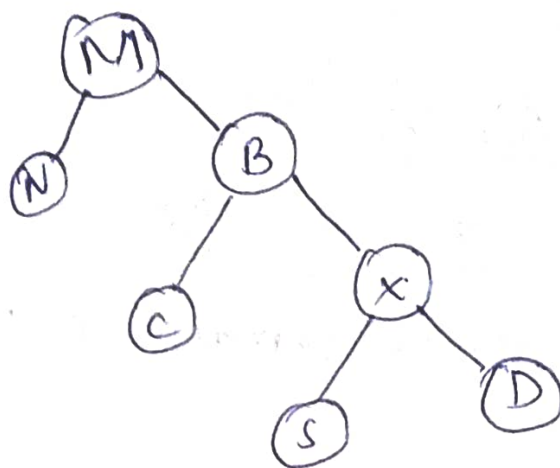
S has no right and left nodes. So the right subtree of x is done.

Step 6: The node before S is C. C is the right node b by seeing the inorder positions.

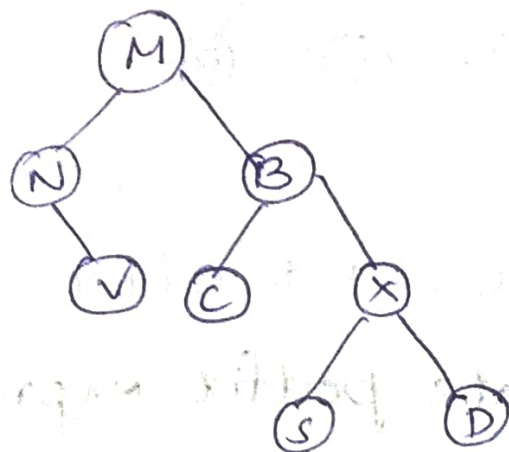


There is no right nodes, remain for root M. So the right subtree is completed.

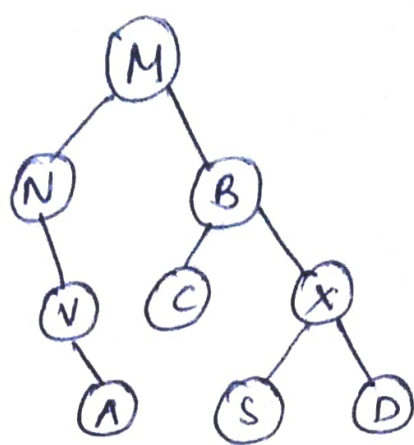
Step 7: The node before c in the preorder is n. n is the left node to the M based on inorder.



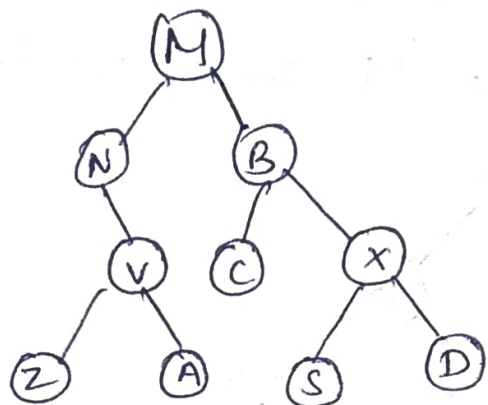
Step 8: The node before n in the postorder is V. The V is right of n in inorder. So V is the right of n.



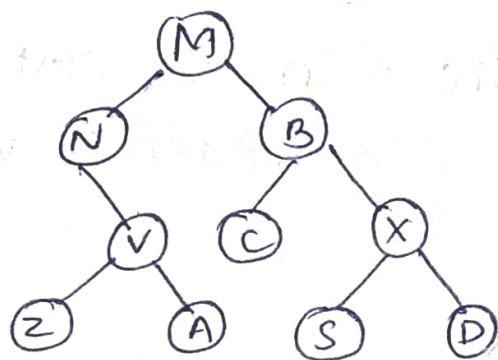
Step 9: The node before V in the postorder is A. A is the right of V in inorder. So A is right of V.



Step 10: The node before A in post order is Z. Z is the left of V in inorder. So, Z is placed left of V.



Step 11: There is no nodes remain in the post order. So the final tree is.



1) Given expression

$$734 + - 245 / + * 6 / 7 +$$

Algorithm of evaluate postfix expression:

A postfix expression is evaluated using a stack.

Rules:

1. Scan the expression from left to right.
2. If the symbol is an operand, push it into the stack.
3. If the symbol is an operator, pop two operands

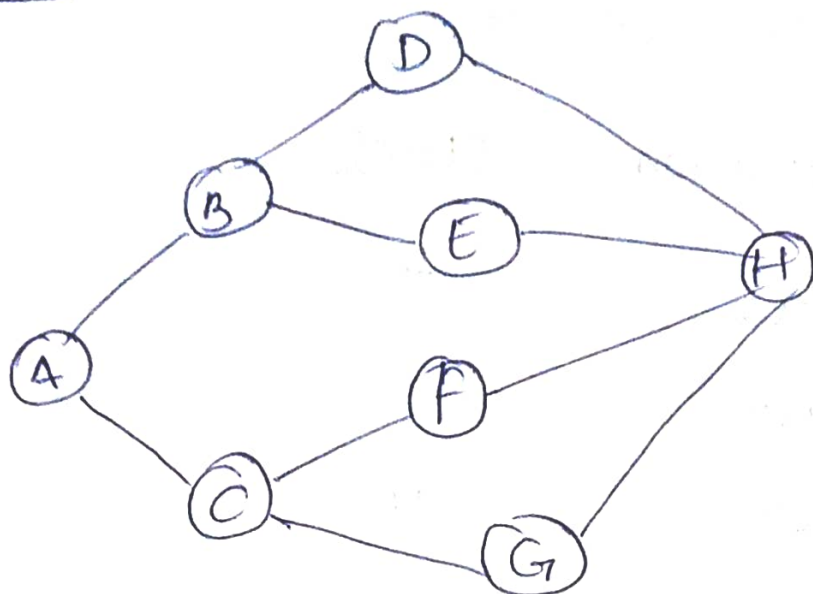
4. Perform the operation and push the result back.
5. Continue until the expression ends.

Read character	Operation	Stack
7	push	7
3	push	3
4	push	4
+	pop(), pop() $4 + 3$	7, 7
-	pop, pop $7 - 7$	0
2	push	0, 2
4	push	0, 2, 4
5	push	0, 2, 4, 5
/	pop(), pop() $4 / 5 = 0.8$	0, 2, 0.8
+	pop(), pop() $2 + 0.8 = 2.8$	0, 2.8
*	pop(), pop() $0 * 2.8 = 0$	0
6	push	0, 6
/	$0 / 6 = 0$	0
7	push	0, 7
+	$0 + 7 = 7$	7

The result is 7

2. DFS:

1)



adj List

A: B, C

B: D, E, A

C: F, G, A

D: H, B

E: H, B

F: H, C

G: H, C

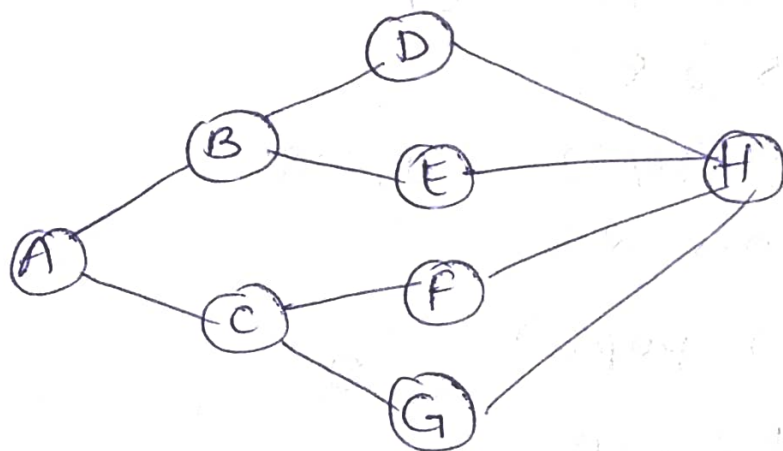
H: D, E, F, G



O/P: A, B, D, H, E, F, C, G

call stack

2) BFS



Adj List

A: B, C

B: D, E, A

C: F, G, A

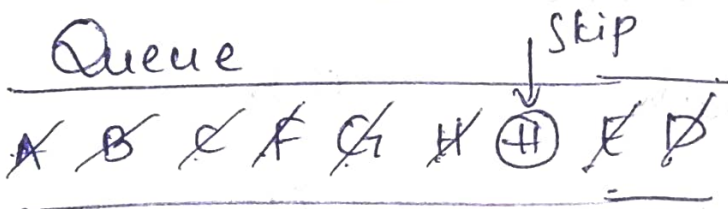
D: H, B

E: H, B

F: H, C

G: H, C

H: G, F, E, D



O/P: A B C F G H E D